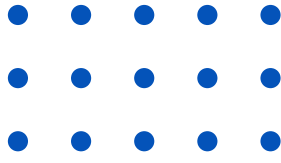


InvisiGuard - Intelligent System for Predicting Cyber Threats Real Time

RP-25-26J-062



Zero-Day Attack Prediction by Analyzing Network Traffic Patterns

Information Technology(Specialization in Cyber Security)

IT22117496

Nawarathna.N.P.D.T



SLIIT
FACULTY OF COMPUTING

IT22117496

Nawarathna.N.P.D.T

RP-25-26J-062

ZERO-DAY ATTACK PREDICTION BY ANALYZING NETWORK TRAFFIC PATTERNS

Existing Problems

- Traditional security systems struggle to detect Zero-Day attacks because they rely on predefined signatures.
- Not only that, but current anomaly-based systems also often fail due to:
 - High volume of false positives.
 - Inability to adapt to new attack patterns.
 - Limited use of machine learning to detect unknown threats.

Uses unsupervised learning model to identify abnormal network behavior.

Predict Unknown Anomaly without relying on the predefined attack signatures.

Give more accurate results based on the latest datasets.



ZERO-DAY ATTACK PREDICTION BY ANALYZING NETWORK TRAFFIC PATTERNS

- **Hybrid ML Architecture**

Uses AE for learn normal traffic and IF for isolate rare patterns.

- **Behavior-Based Security Design**

Focuses on traffic patterns rather than known attack indicators.

- **Reduced False Positives**

Layered analysis filters noise before deep classification.

- **Real-Time & Scalable**

Ability to monitor modern network environments.

System provides useful insights even without labeled attack data

Reduced false positives compared to rule-based detection

Severity tagging helps users to prioritize incidents faster.



Predicting Known Insider Attacks Using User Behavioral Patterns

Information Technology(Specialization in Cyber Security)

IT22133304

Pallewaththa P.A



SLIIT
FACULTY OF COMPUTING

IT22133304

Pallewaththa P.A

RP-25-26J-062

PREDICTING KNOWN INSIDER ATTACKS USING USER BEHAVIOURAL PATTERNS

Traditional security systems struggle to accurately model user behavior in large enterprises, often missing subtle anomalies in login activity, file access, and USB usage due to reliance on static rules and isolated events.

The proposed user behavior analysis component of InvisiGuard addresses this by:

The solution combines multiple user and system behavioral signals using advanced deep learning models such as LSTM and CNN, enabling the construction of adaptive and evolving behavioral baselines. By continuously learning from both long-term behavior patterns and short-term activity anomalies, the system delivers real-time, interactive risk visualizations that support faster, more informed security decision-making.



PREDICTING KNOWN INSIDER ATTACKS USING USER BEHAVIOURAL PATTERNS

✨ Hybrid Deep Learning Architecture

Combines LSTM for modeling long-term user behavior sequences with CNN for detecting short-term abnormal activity bursts.

Captures both temporal trends and localized anomalies in insider behavior.

🧠 Personalized Behavioral Baselines

Builds individual user behavior profiles instead of generic rules.

Adapts detection to different roles, access levels, and working patterns, reducing false positives.

🔗 Unified Behavioral Intelligence

Integrates login patterns, file access, after-hours activity, and USB usage into a single detection framework.

Overcomes limitations of isolated or rule-based analysis.

⚡ Real-Time & Scalable Detection

Designed for continuous monitoring of high-volume enterprise logs.

Ensures low-latency alerts while remaining scalable for large organizations.

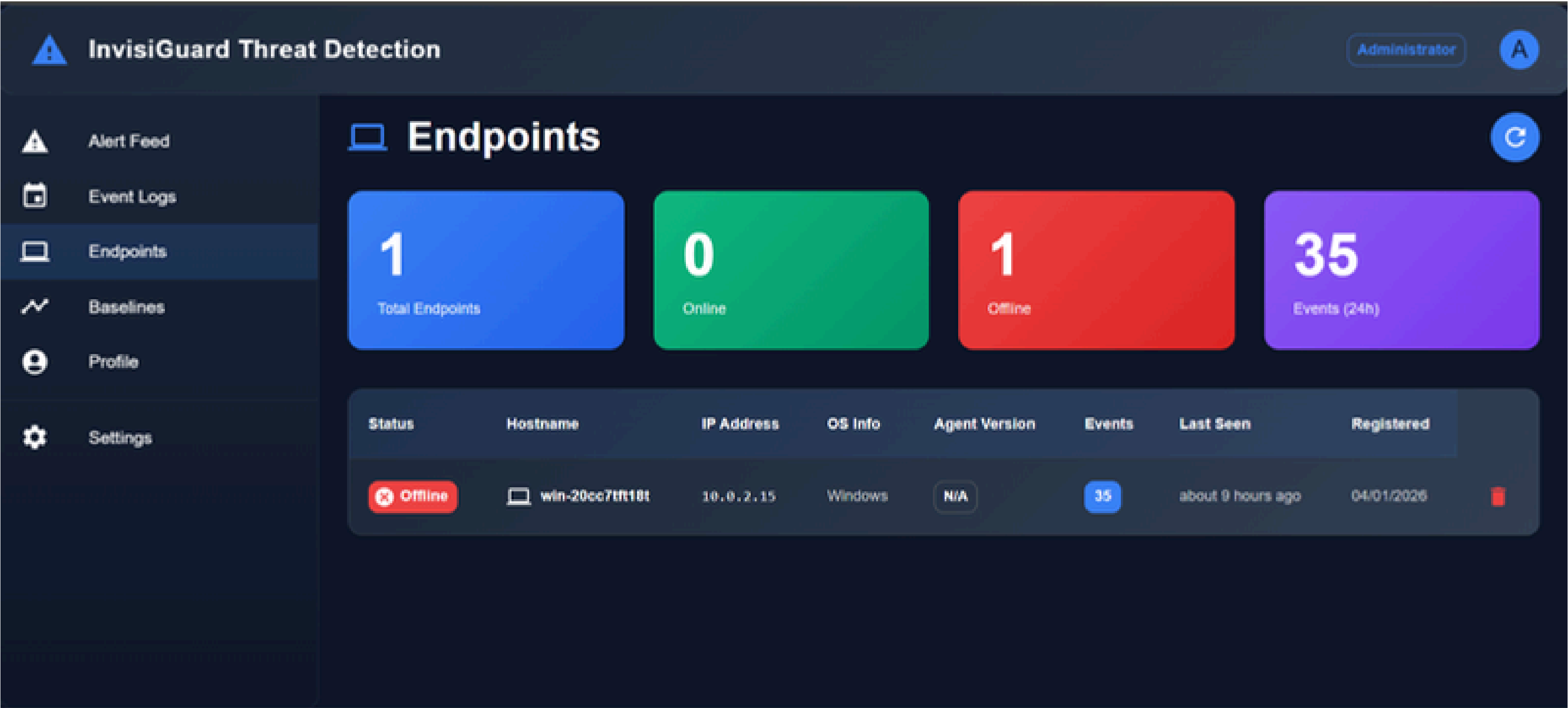
Accurate prediction of insider threats based on real user behavior

Reduced false positives compared to rule-based detection

Support for diverse user activities (logins, files, devices)

Analyst-friendly outputs for investigation and decision-making

Progress As PP1



Known Attack Prediction by Analyzing Endpoint Log Patterns

Information Technology(Specialization in Cyber Security)

IT22562906 Nethkalum G M H



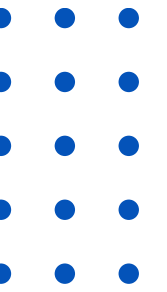
SLIIT
FACULTY OF COMPUTING

IT22562906

Nethkalum G M H

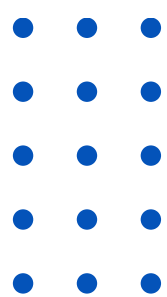
RP-25-26J-062

Known Attack Prediction by Analyzing Endpoint Log Patterns



In today's dynamic cybersecurity environment, traditional signature-based and rule based intrusion detection systems struggle against sophisticated attacks and advanced persistent threats. The sheer volume of endpoint devices and log data creates an urgent need for intelligent, scalable security solutions.

The solution combines unified endpoint log analysis with NLP-based feature engineering and a two-stage ML architecture. Raw Windows are transformed using TF-IDF and analyzed through a fast One-Class SVM filter followed by XGBoost classification, enabling accurate and real-time detection of known cyberattacks.



Known Attack Prediction by Analyzing Endpoint Log Patterns

✨ **Hybrid ML Architecture**
Uses SVM for fast anomaly filtering and XGBoost for precise attack prediction.

🧠 **Context-Aware Endpoint Analysis**
Learns normal behavior from process execution .

🧩 **Reduced False Positives**
Layered analysis filters noise before deep classification.

⚡ **Real-Time & Scalable**
Supports low-latency monitoring of enterprise endpoint logs.

Early detection of unknown endpoint threats

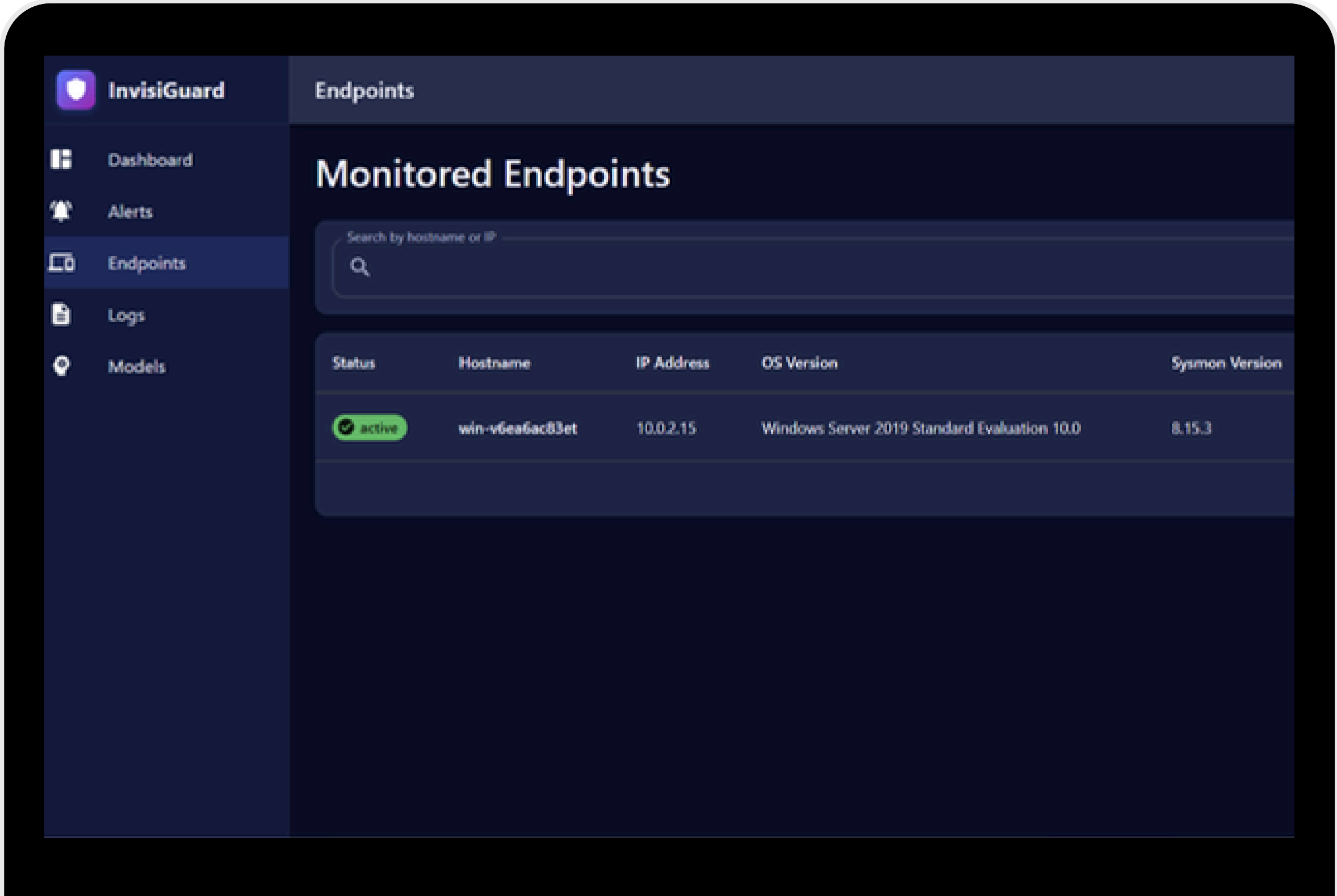
Reduced reliance on predefined attack signatures

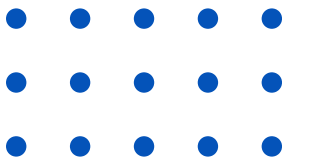
Real-time monitoring of endpoint activity

Clear visibility into suspicious endpoint actions



Progress As of PP1





KNOWN ATTACK PREDICTION BY ANALYZING SYSTEM CALL PATTERNS

Information Technology(Specialization in Cyber Security)

IT22343116 Nawarathna M I M N



SLIIT
FACULTY OF COMPUTING

IT22343116

Nawarathna M I M N

RP-25-26J-062

KNOWN ATTACK PREDICTION BY ANALYZING SYSTEM CALL PATTERNS.

Sub-Problem

Traditional intrusion detection systems struggle to analyze host-level behavior, especially system-call patterns, in a way that captures both statistical activity and temporal execution flow.

Our Solution

The proposed system-call component of InvisiGuard addresses this by:

Analyzing system-call behavior from endpoint execution traces.

Using machine learning models to learn known attack patterns.

Producing a risk-based suspicious level instead of a rigid binary decision.



KNOWN ATTACK PREDICTION BY ANALYZING SYSTEM CALL PATTERNS.

Design Excellence & Contribution

Hybrid ML Architecture

Random Forest → statistical behavior analysis
LSTM → sequential system-call pattern learning

Risk-based output model

Outputs suspicious levels (Low / Medium / High)

Modular & extensible design

Each component (RF, LSTM, hybrid logic) can be independently improved

Windows-focused host-based detection

Addresses a gap where most public datasets are Linux-centric

Novelty

Combining **statistical + sequential** learning at the **system-call level** with confidence-based hybrid scoring.

User Feedback on Prototype

Clear visibility into attack confidence levels
Better understanding of behavioral risk, not just alerts
Useful foundation for integration into a dashboard or EDR-style tool
Positive feedback on explainability and staged detection logic

Future user feedback will focus on

Benign behavior integration
Alert threshold tuning





InvisiGuard – Intelligent System for Predicting Cyber Threats in Real Time



1. Customer Persona

- ✓ SOC Analysts
- ✓ Enterprise IT Security Teams
- ✓ CISOs & Risk Management Units
- ✓ Large & Mid-scale Organizations

✓ **Who benefits:** Organizations handling large-scale logs and sensitive data



2. Market Size



- **Global cybersecurity market** growing rapidly
- High demand for **AI-based threat prediction**
- Increasing focus on **zero-day & insider threat detection**
- **Target sectors:** Finance, Telecom, Healthcare, Government

✓ **Strong demand for proactive security solutions**

3. What Is Unique in InvisiGuard



- **Unified 4-component AI framework**
- Predicts known + **zero-day** attacks
- **Hybrid ML & Deep Learning** models
- **Multi-source analysis:** System Calls, User Behavior, Network Traffic, Endpoint Logs

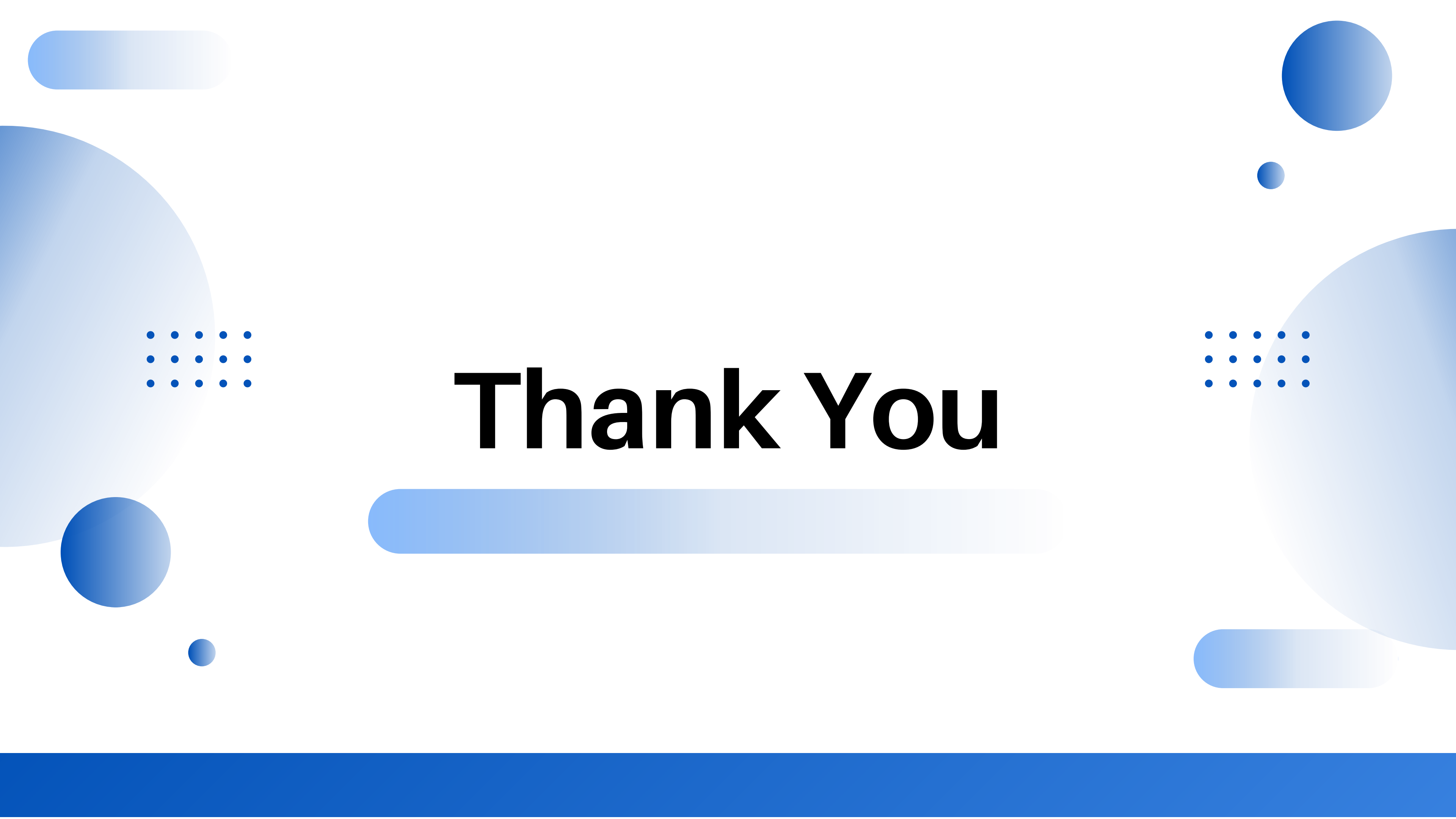
✓ **Moves from reactive detection → predictive defense**

4. Cost Recovery / Sustainability



- **Subscription-based SaaS model**
- Enterprise licensing
- Optional premium analytics & dashboards
- **Scalable cloud** deployment
- **Low operational cost** using open-source ML

✓ **Sustainable, scalable & commercially viable**

The background features a minimalist abstract design. On the left, a large light blue circle is partially visible, with a smaller solid blue circle and a tiny blue dot below it. A horizontal rounded rectangle is at the top left, and a 3x5 grid of small blue dots is positioned to the right of the large circle. On the right, another large light blue circle is partially visible, with a solid blue circle and a tiny blue dot above it. A horizontal rounded rectangle is at the bottom right, and a 3x5 grid of small blue dots is positioned to the left of the large circle. A thick blue horizontal bar spans the entire width of the image at the bottom.

Thank You